DoTA: Latent Distribution Conditioned Data Attribution for Diffusion Models

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Motivation

- Diffusion models are trained on large-scale copyrighted artistic datasets such as LIOAN **[8]** and can regenerate stylistic or structural patterns from training data.
- Identifying which training images influence a generated output is essential for intellectual property analysis, model auditing, and ensuring ethical and legal accountability.
- Existing data attribution methods **[4-7]** are inadequate for diffusion models due to stochastic denoising dynamics, non-convex optimization, high computational cost, and strong architectural dependence.

Illustration

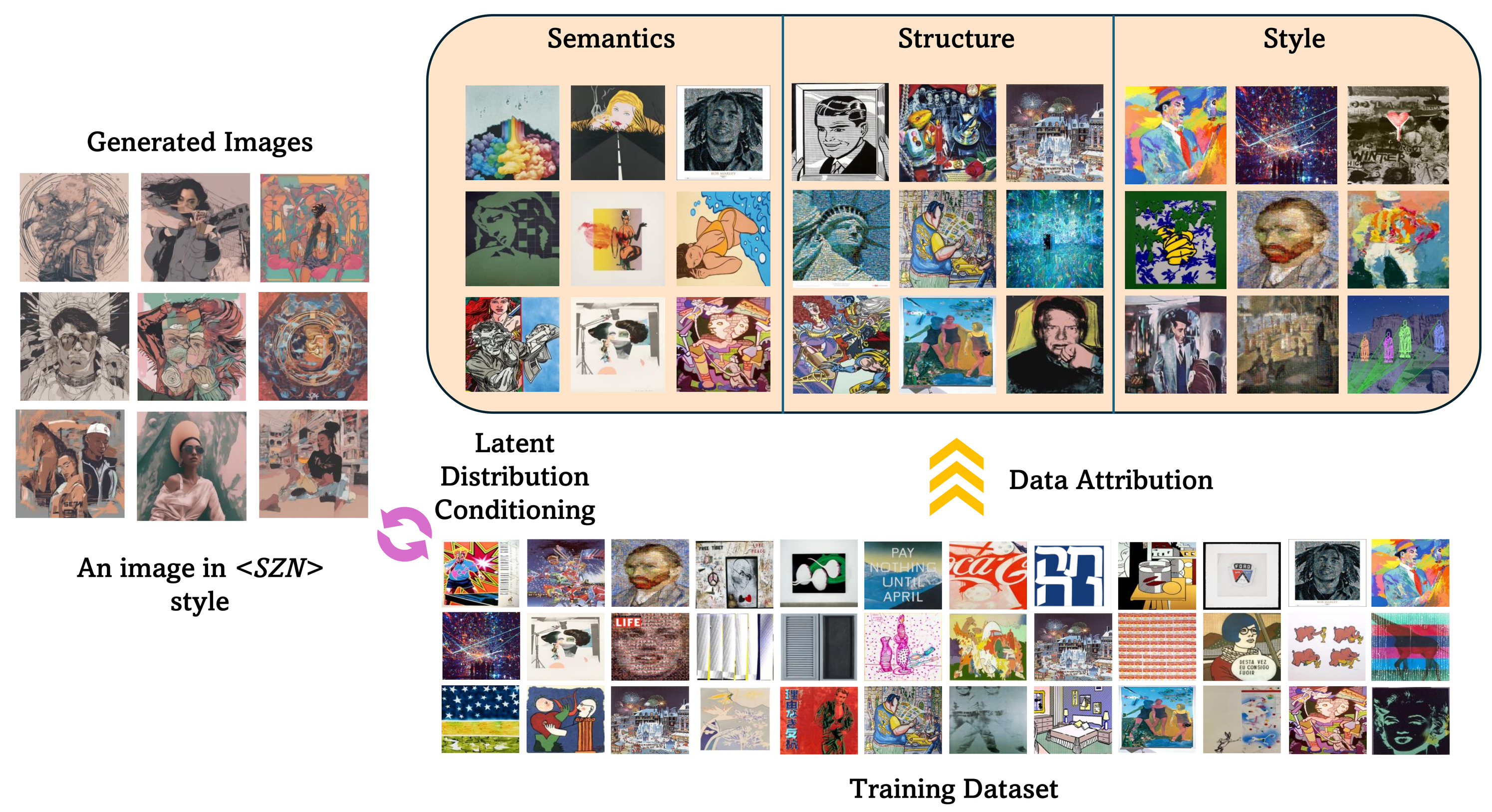

Figure 1: We present a data attribution method for text-to-image diffusion models using latent distribution conditioning for fine-grained control. We analyze attribution across three dimensions: semantics, structure, and style. Semantics captures object-level concepts, structure models spatial arrangement and pose, and style reflects visual characteristics such as texture, color palette, and artistic rendering. Target images are generated using a fine-tuned diffusion model, where the prompt includes a unique identifier, <SZN>, to consistently invoke the desired style (e.g., PopArt).

Latent Representation for Attribution

- **Semantics:** CLIP **[1]** embeddings are used to capture high-level conceptual similarity.
- **Structure:** DINOv2 **[2]** features are used to model spatial layout and pose information.
- **Style:** Gram matrix **[3]** representations are used to capture texture and stylistic statistics.

Data Attribution via DoTA

- **Latent Distribution–Conditioned Attribution.** We define attribution in latent space, where influence is measured by distributional alignment between training subsets and a generated sample. Operating in latent representations gives us the freedom to choose the attribute of interest (e.g., semantic, structural, stylistic), making attribution attribute-specific and controllable.
- **Fine-Tuning for Attribute Conditioning.** We optionally fine-tune the diffusion model on an attribute-focused dataset to ensure that the chosen latent representation meaningfully captures the target artistic factor, enabling faithful downstream attribution.
- **Search Space Pruning via Energy Distance (ED).** Full Shapley evaluation over all training samples is computationally prohibitive. We construct coverage-controlled subsets and compare their latent distributions to the generated sample using **Energy Distance (ED)**. Samples consistently appearing in low-ED subsets are retained, reducing the candidate pool while preserving statistically aligned contributors. Working in latent space makes this comparison efficient and representation-aware.
- **Shapley Values with ED Utility.** Training samples are treated as cooperative players and Shapley values are computed using **KernelSHAP [10]**. The utility function is defined via negative **Energy Distance (ED)**, ensuring attribution scores reflect the chosen latent attribute. Using ED as the utility guarantees that Shapley values are grounded in distributional alignment rather than gradient heuristics.

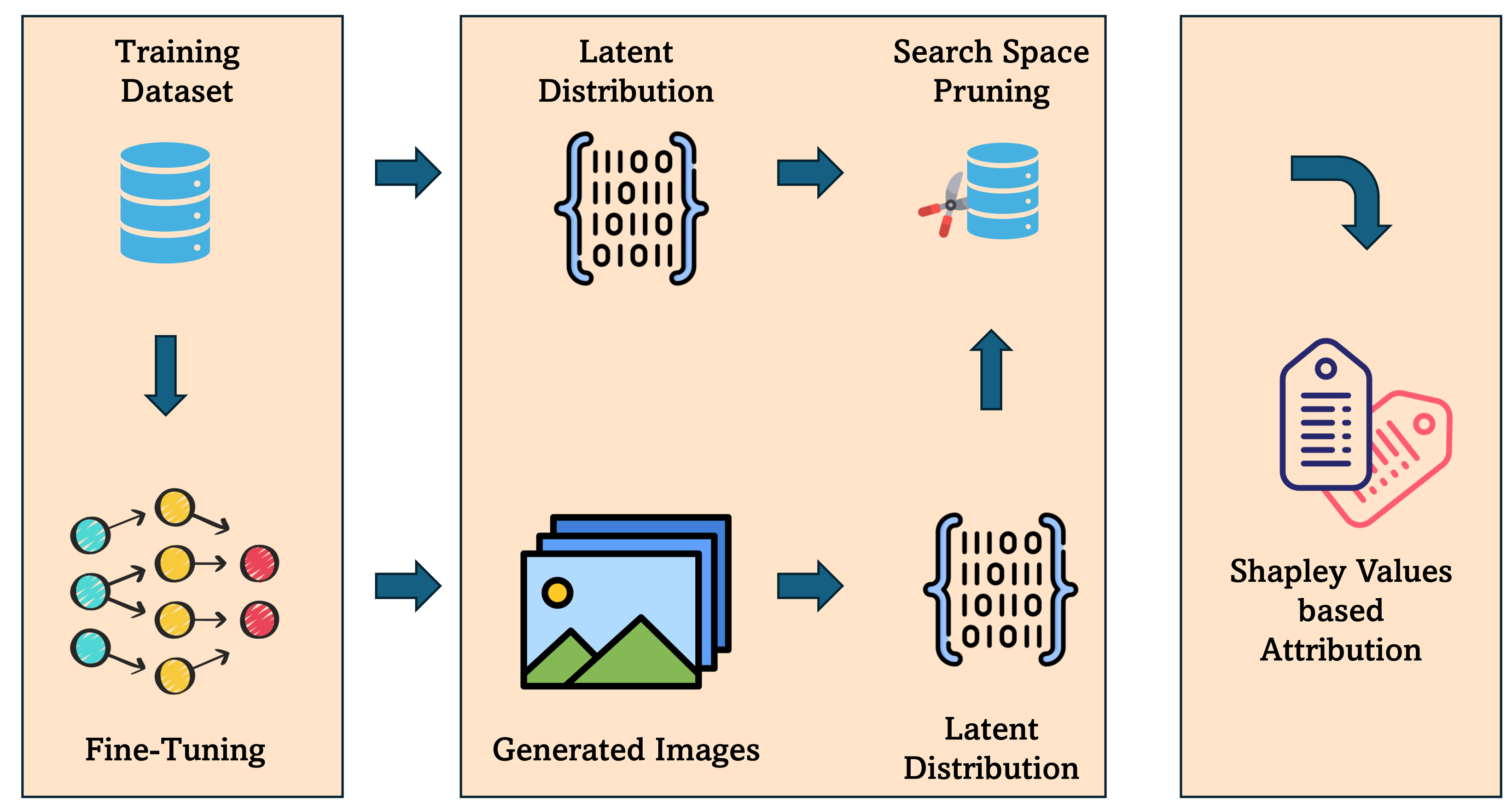

Figure 2: Architecture diagram of our proposed data attribution technique *DoTA*. We attribute the generated images on the images in the training dataset. The images are generated with a fine-tuned diffusion model on the training dataset. The latent distribution conditioned search space pruning of the training dataset allows to reduce the noise in the candidates participating in the Shapley Values-based attribution.

Experiments & Results

- **Datasets.** Fine-tuning and evaluation on PopArt (*WikiArt* **[9]**), curated *Ukiyo-e* collections **[12]**, and the *Munch* dataset **[13]** to test attribution across semantic, structural, and stylistic latent factors under controlled artistic variation.
- **Counterfactual Faithfulness and Metrics.** We evaluate attribution using Leave-K and Take-K retraining. In Leave-K, removing top-K attributed samples should reduce alignment if attribution is faithful; in Take-K, training only on top-K samples should preserve alignment. Performance is measured using a style classifier trained on **UnlearnCanvas [11]**. **ASC** (Absolute Style Confidence) is defined as the classifier probability assigned to the target style for the generated image, and **RSC** (Relative Style Confidence) is defined as the improvement in classifier probability over a random subset baseline.
- **Adversarial Setup.** We inject adversarial artworks into the attribution pool and evaluate top-K ranking stability along with changes in **ASC** and **RSC**. DoTA remains robust due to Energy Distance (ED)–based distributional alignment.
- **Efficiency.** We report runtime and memory for Full Shapley estimation, DoTA without pruning, and DoTA with ED-based Search Space Pruning, demonstrating significant computational reduction while preserving attribution fidelity.

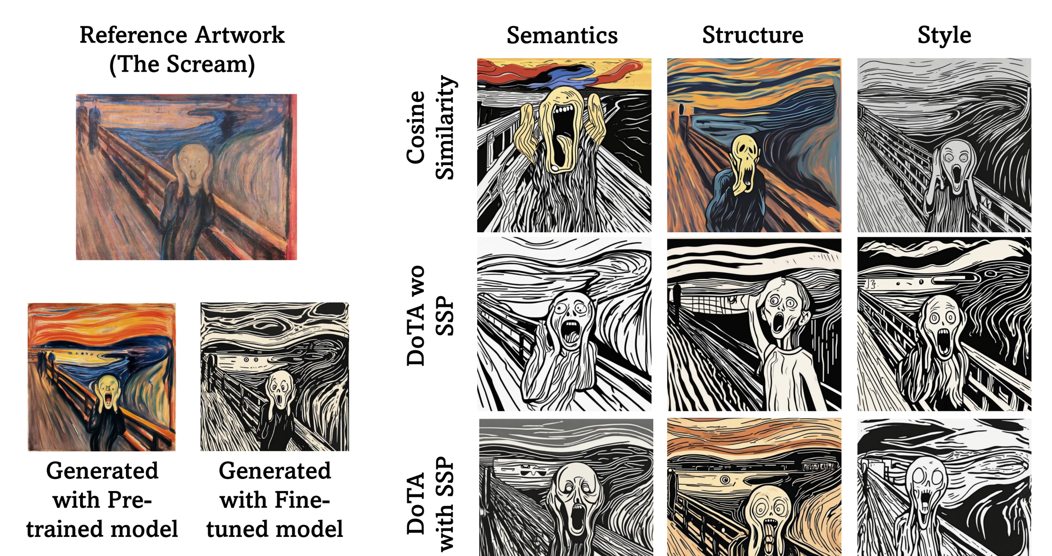

Figure 3: Qualitative comparison of attribution approaches under adversarial attack with different latent representations for the PopArt dataset. We present the **images generated for "The Scream" after retraining with Take-K under adversarial attack** in different settings. We observe maximum impact of the attack with the cosine similarity based attribution showing drastic changes in semantics, structure, and style in the respective generated images.

References

1. CLIP (ICML, 2021)
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3. NeuralStyle (CVPR, 2016)
4. TRAK (ICML, 2023)
5. DTRAK (arXiv, 2024)
6. JTRAK (arXiv, 2024)
7. InfluenceFunctions (ICML, 2017)
8. LIOAN (NeurIPS, 2023)
9. WikiArt (arXiv, 2016)
10. SHAP (NeurIPS, 2017)
11. UnlearnCanvas (NeurIPS, 2023)
12. Ukiyoe (CVPR, 2018)
13. Munch (WikiArt, 2016)